

Secure Diagnostic Laboratory Network

Enterprise Network Architecture & Security Hardening for Multi-Branch
Diagnostic Data Transmission

- Cisco Packet Tracer v8.x

- Extended ACLs & NAT/PAT

- VLAN Segmentation

- SSHv2 Hardening

- WPA2-Enterprise / WLC

- HTTPS/TLS Encrypted Transit

- Python Automation

- Wireshark Traffic Audit

PROJECT TYPE & ROLE

Network Implementer & Simulation Lead

Group Diploma Project (4 Members)

IMPLEMENTATION TIMELINE

February 2026 – June 2026

Simulation, Hardening & Audit

DEPLOYMENT CLIENT / TARGET

AVM Labs (Qurum, Muscat)

17 Interconnected Clinic Branches

VERIFICATION EFFICACY

18 / 18 Test Cases Passed (100%)

Negative & Positive Validation

01. Executive Summary & Network Scenario

**Clinical Operational Context**

AVM Labs operates as a premier clinical diagnostic center located in Qurum, Muscat. The core infrastructure processes pathology, biochemistry, and imaging records transmitted continuously across **17 regional partner clinics**.

**Threat Vector & Identified Problems**

Prior to this project, data exchanges operated across flat, unsegmented subnets with unhardened Telnet and cleartext transfers—exposing sensitive patient data to eavesdropping and lateral pivoting.

17
CLINIC BRANCHES

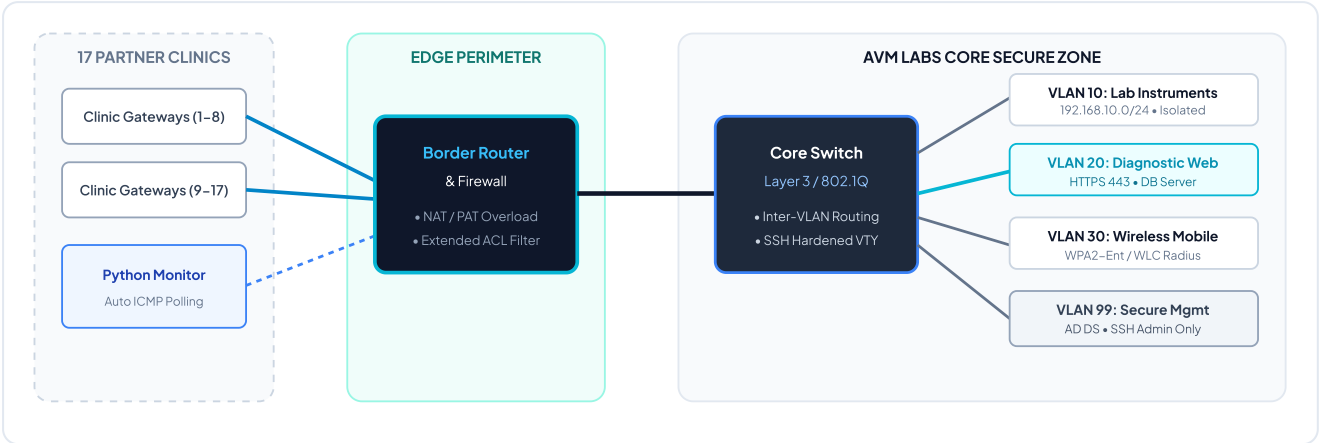
4
DEDICATED VLANs

18/18
TESTS PASSED

100%
SSHv2 HARDENED

02. Enterprise Architecture & Traffic Flow

A defense-in-depth architecture was simulated in **Cisco Packet Tracer**. Logical segmentation isolates operational zones, while perimeter firewalling and Active Directory services enforce least privilege.



Layered Defense-in-Depth Control Matrix

Security Domain	Implementation Mechanism	Technical Objective
Segmentation	VLANs (10, 20, 30, 99) & IEEE 802.1Q	Isolates lab hardware, wireless clients, and servers into bounded subnets.
Access Control	Extended Access Control Lists (ACLs)	Enforces top-down filtering; restricts database access strictly to HTTPS 443.
Edge Boundary	Cisco IOS Firewall Policies & NAT/PAT	Translates private RFC 1918 space and drops unauthorized inbound probes.
Administration	SSHv2 (RSA 2048-bit), Telnet Blocked	Guarantees cryptographic confidentiality for network administrator sessions.
Wireless / Identity	WPA2-Enterprise (802.1X / RADIUS / AD DS)	Replaces pre-shared keys with centralized credential lifecycle verification.
Data in Transit	HTTPS / TLS Session Encryption	Shields medical diagnostic reports against man-in-the-middle snooping.

03. Individual Role & Key Contributions

Role: Network Implementer & Simulation Lead

As technical lead for network implementation, I was personally responsible for designing the Cisco Packet Tracer simulated topology, engineering the Access Control Lists, hardening device lines with SSHv2, resolving trunking faults, and building the Python connectivity monitoring utility.

Core Implementation Scope

- ✓ Engineered 802.1Q trunking and sub-interface routing logic.
- ✓ Authored Extended ACLs preventing lateral pivot across subnets.
- ✓ Hardened VTY lines with SSHv2 and 2048-bit RSA keys.
- ✓ Configured perimeter NAT/PAT overload pools for clinic endpoints.

Quality & Verification Focus

- ✓ Constructed standalone testbed for script validation.
- ✓ Conducted Wireshark packet inspections to audit packet drops.
- ✓ Developed automated Python node telemetry alert script.
- ✓ Authored individual troubleshooting logs and risk matrix.

04. Technical Hardening & Code Implementations

1. Access Control Policy & Cisco IOS CLI Implementation

Extended ACLs were constructed using top-down sequential evaluation. General access is denied by default, while explicit HTTPS channels (TCP 443) to the Diagnostic Server (192.168.20.10) are permitted from clinic segments.

Cisco IOS CLI - Extended ACL Hardening Configuration

Cisco Packet Tracer

```
! Define Secure Diagnostic Access Control List
ip access-list extended SECURE_DIAGNOSTIC_ACL
  remark === Permit Encrypted Clinic Web Traffic to Diagnostic Web Server ===
  permit tcp 192.168.10.0 0.0.0.255 host 192.168.20.10 eq 443
  permit tcp 172.16.0.0 0.0.255.255 host 192.168.20.10 eq 443

  remark === Block All Other Direct IP Access to Diagnostic Host ===
  deny ip any host 192.168.20.10 log

  remark === Allow Outbound Lab Subnet Traffic & Block Lateral Movement ===
  permit ip 192.168.10.0 0.0.0.255 any
  deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
  deny ip any any
```

2. Management Plane Hardening (SSHv2 vs. Telnet)

SSHv2 Implementation

Configured `ip ssh version 2` with 2048-bit modulus RSA keys. VTY lines configured with `transport input ssh`, `login local`, and an inactivity timer of `exec-timeout 5 0` to prevent session hijacking.

Complete Telnet Removal

Unencrypted TCP port 23 communication was eliminated across all switches and routers. Packet captures confirmed cleartext credential transmission is fully mitigated.

05. Clinic Telemetry Monitor (Python Automation)

To detect connectivity failures across the 17 clinic branches without manual polling, an automated ICMP health-check script was developed and validated in a dedicated Packet Tracer test bench.

Python 3 - clinic_node_monitor.pyAutomated Health Monitor

```
import subprocess, platform, time
from datetime import datetime

CLINIC_NODES = {
    "Clinic_01_Muscat": "10.0.1.1",
    "Clinic_02_Muttrah": "10.0.2.1",
    "Diagnostic_Server": "192.168.20.10"
}

def ping_node(host: str) -> bool:
    flag = "-n" if platform.system().lower() == "windows" else "-c"
    cmd = ["ping", flag, "1", "-W", "2", host]
    return subprocess.run(cmd, stdout=subprocess.DEVNULL, stderr=subprocess.DEVNULL).returncode == 0

def run_health_sweep():
    print(f"[*] Telemetry Sweep: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}")
    for name, ip in CLINIC_NODES.items():
        status = "ONLINE [OK]" if ping_node(ip) else "OFFLINE [ALERT!]"
        print(f" -> Node: {name:<20} | IP: {ip:<15} | Status: {status}")

if __name__ == "__main__":
    run_health_sweep()
```

06. Troubleshooting & Engineering Challenges

#	Challenge Encountered	Root Cause Analysis	Resolution Applied
1	VLANs unable to reach WAN	Switch uplink port remained in mode access ; 802.1Q missing.	Configured switchport mode trunk on uplinks & verified dot1Q encapsulation.
2	ACL permitted unauthorized traffic	Rule sequence error: generic permit preceded specific deny .	Restructured ACL sequence using IOS line numbers to enforce top-down match logic.
3	Legitimate clinic requests dropped	Inverted wildcard mask calculation (matched incorrect host range).	Recalculated inverse masks (e.g., 0.0.0.255 for /24) & validated.
4	Group topology bottleneck	Script verification blocked while main topology was being integrated.	Constructed an independent, mini test bench to validate script logic in parallel.

07. Verification & Testing Battery

18/18 Test Cases Validated (100% Pass)

Rigorous functional and security testing was executed across VLAN boundaries, NAT translation tables, SSH management, WPA2-Enterprise authentication, and HTTPS web sessions.

Negative Testing Philosophy:

"Verifying connectivity proves a network functions; verifying that unauthorized packets are explicitly dropped proves a network is secure."

08. Key Technical Takeaways & Lessons Learned

💡 Technical Insights

- **ACL Order is Policy:** First-match evaluation means a single misplaced allow rule completely voids perimeter security.
- **Default States Lie:** Never assume switchport negotiation or native VLAN matches—always explicitly define trunking modes.
- **PDU Tracing:** Simulation mode in Cisco Packet Tracer was indispensable for tracking TCP handshake completion across sub-interfaces.

👥 Project & Teamwork Insights

- **Version Control for .pkt Files:** Standardized file naming (`topology_v1.2_acl.pkt`) avoided team configuration overwrites.
- **Single Source IP Matrix:** Maintaining a shared subnet and gateway sheet cut debugging turnaround time in half.
- **Modular Validation:** Validating subnets individually prior to full-mesh convergence minimized cascading errors.

09. Engineering Limitations & Future Roadmap

Simulation Constraints

- **Simulation Boundary:** Cisco Packet Tracer emulates software behavior; real hardware exhibits ASIC-level throughput constraints.
- **WAN Emulation:** Branch clinics used static routing rather than production-grade dynamic routing overlays.

Production Roadmap

- ✓ **OSPF Multi-Area:** Deploy dynamic routing across clinic routers.
- ✓ **IPsec VPN Tunnels:** AES-256 site-to-site WAN encryption.
- ✓ **Gateway Redundancy:** Implement HSRP / VRRP high availability.
- ✓ **Intrusion Detection:** Integrate Snort/Suricata sensor monitoring.

10. Portfolio & Academic Attribution

Academic Context: This portfolio summary represents a sanitized, public-facing extract of an academic group diploma project completed between February 2026 and June 2026.

Individual Attribution: This document highlights my personal contributions as the **Network Implementer & Simulation Lead** (topology architecture, Extended ACL engineering, SSHv2 hardening, and Python automation). It does not claim sole authorship of the broader collaborative group project.

Sanitization Note: All institutional student identifiers, personal signatures, confidential passwords, and formal evaluation rubrics have been omitted in compliance with academic privacy and security best practices.

DESIGNED & ENGINEERED FOR ENTERPRISE NETWORK SECURITY

Cisco Packet Tracer • Wireshark • Python • Active Directory